15. Analyse various types of servers using Webalizer tool.

Experiment No. 15

Experiment Name

Analysis of Various Types of Servers Using Webalizer Tool

Aim

To analyze web server access logs using the Webalizer tool and study the statistical information generated from server activity. To understand how web server logs can be used to identify the number of requests, visitors, accessed pages, data transfer, and other useful information about website usage.

Procedure

Part A: Setting Up the Web Server

1. Open the XAMPP Control Panel on the computer. XAMPP provides an Apache web server that can be used to generate access logs for analysis.
2. Start the Apache service by clicking the **Start** button next to Apache in the XAMPP Control Panel. Wait until the service starts successfully. If Apache does not start, check whether another application is using its configured port.
3. Open a web browser and enter http://localhost/ in the address bar. The local web server page should appear if Apache is running correctly.
4. Access the local web server several times by refreshing the page and opening different available local pages. These requests generate entries in the Apache access log.
5. Locate the Apache access log file. In a standard XAMPP installation, the file is usually available at:

C:\xampp\apache\logs\access.log

6. Open the access log file using a text editor. Examine the recorded entries, which commonly include the client's IP address, request time, requested resource, HTTP status code, and response size. The exact information depends on the server's log configuration.

Part B: Installing and Configuring Webalizer

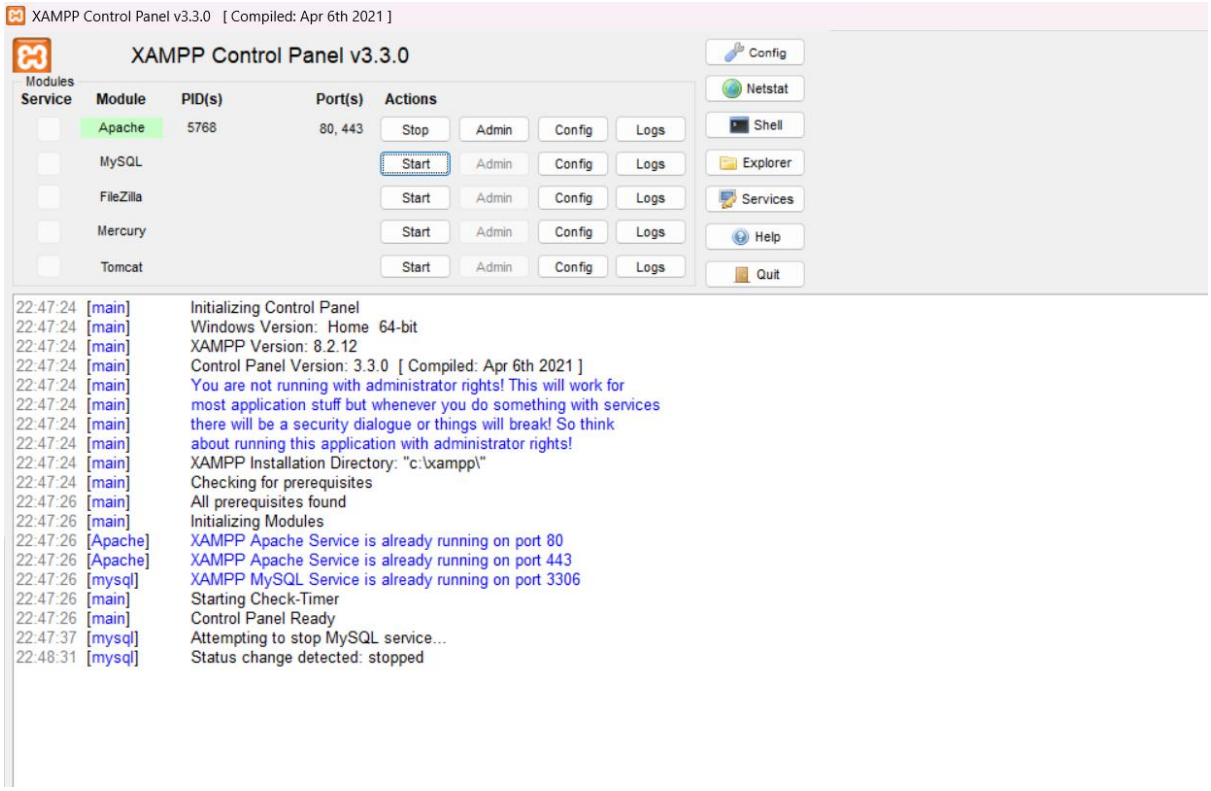
7. Obtain a compatible Webalizer distribution for the operating system from a trusted source. Webalizer is a separate log-analysis utility; it is not the same as the Apache server and is not automatically configured simply by installing XAMPP.
8. Extract or install Webalizer according to the instructions provided with the selected distribution. Ensure that the executable and its configuration files are accessible from the computer.
9. Configure Webalizer to read the Apache access log generated by XAMPP. Specify the correct path to the log file and configure the output directory where the generated reports will be stored.
10. Review the configuration options and ensure that the log format selected by Webalizer is compatible with the format used by Apache. If the formats differ, adjust the configuration before processing the log.
11. Run Webalizer using its supported command-line options to process the access log. The exact command depends on the installed version and its configuration. Check the program's documentation if the executable requires additional arguments.
12. Verify that Webalizer processes the log without errors. If an error occurs, check the input log path, file permissions, log format, configuration file, and output directory.

Part C: Generating and Analyzing Reports

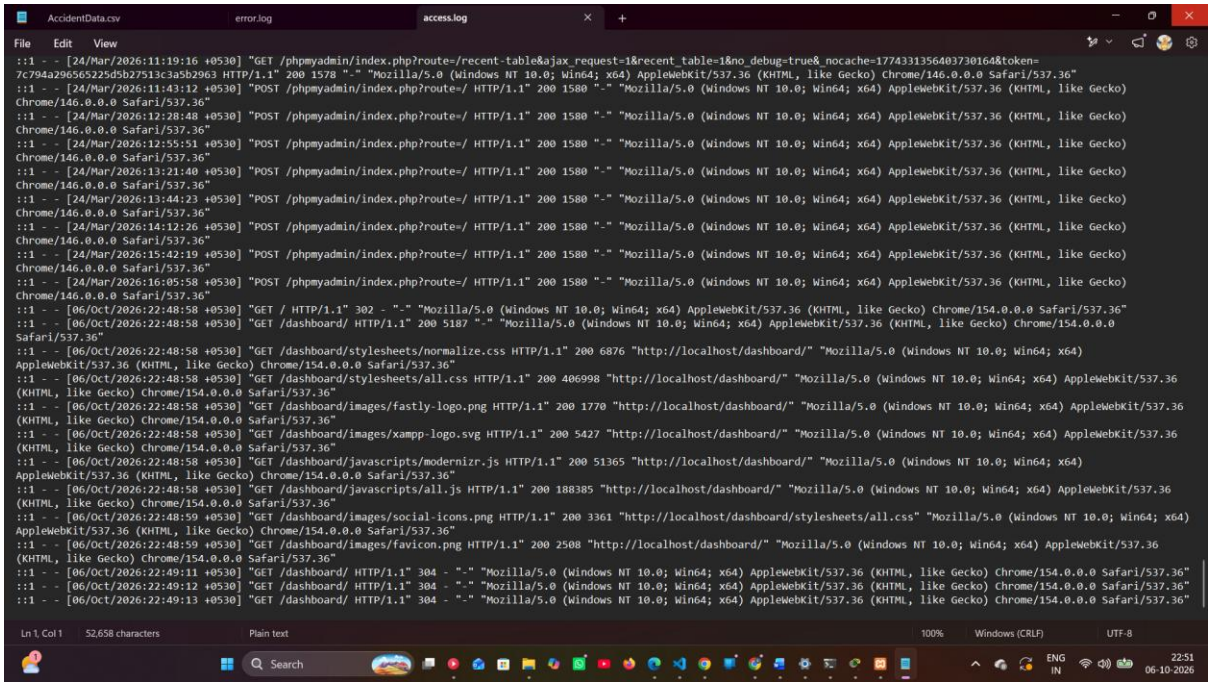
13. Open the output directory configured for Webalizer. When processing is successful, the utility generates HTML-based statistical reports that can be viewed in a web browser.
14. Open the generated report's main HTML page in a browser. Examine the summary of web server activity for the period covered by the available log data.
15. Study the statistics displayed in the report. Depending on the log contents and Webalizer configuration, these may include the number of requests, visits, unique sites, requested URLs, transferred data, and response codes.
16. Examine the monthly summary and other available breakdowns to understand how server activity changes over time. If only a small number of local requests have been generated, the report will contain correspondingly limited statistics.
17. Identify the most frequently requested resources from the URL or page statistics. This helps determine which pages or files were accessed most often during the analyzed period.
18. Examine the status-code statistics, if available. HTTP status codes provide information about request outcomes. For example, status code 200 generally

indicates success, while status code 404 indicates that the requested resource was not found.

19. Compare the generated statistics with the original Apache access log. Verify that the report reflects the requests made while accessing the local server. Keep in mind that a request count is not necessarily the same as the number of individual visitors.
20. Finally, record the observations from the report and explain how server-log analysis can help administrators understand usage patterns, identify frequently accessed resources, and monitor server activity.


XAMPP Control Panel v3.3.0 [Compiled: Apr 6th 2021]
XAMPP Control Panel v3.3.0
Modules
Service Module PID(s) Port(s) Actions
Apache 5768 80, 443 Stop Admin Config Logs
MySQL Start Admin Config Logs
FileZilla Start Admin Config Logs
Mercury Start Admin Config Logs
Tomcat Start Admin Config Logs
Config
Netstat
Shell
Explorer
Services
Help
Quit
22:47:24 [main] Initializing Control Panel
22:47:24 [main] Windows Version: Home 64-bit
22:47:24 [main] XAMPP Version: 8.2.12
22:47:24 [main] Control Panel Version: 3.3.0 [Compiled: Apr 6th 2021]
22:47:24 [main] You are not running with administrator rights! This will work for
22:47:24 [main] most application stuff but whenever you do something with services
22:47:24 [main] there will be a security dialogue or things will break! So think
22:47:24 [main] about running this application with administrator rights!
22:47:24 [main] XAMPP Installation Directory: "c:\xampp\"
22:47:24 [main] Checking for prerequisites
22:47:26 [main] All prerequisites found
22:47:26 [main] Initializing Modules
22:47:26 [Apache] XAMPP Apache Service is already running on port 80
22:47:26 [Apache] XAMPP Apache Service is already running on port 443
22:47:26 [mysql] XAMPP MySQL Service is already running on port 3306
22:47:26 [main] Starting Check-Timer
22:47:26 [main] Control Panel Ready
22:47:37 [mysql] Attempting to stop MySQL service...
22:48:31 [mysql] Status change detected: stopped


localhost/dashboard/
School
Apache Friends FAQs HOW-TO Guides PHPInfo phpMyAdmin
XAMPP Apache + MariaDB + PHP + Perl
Welcome to XAMPP for Windows 8.2.12
You have successfully installed XAMPP on this system! Now you can start using Apache, MariaDB, PHP and other components. You can find more info in the FAQs section or check the HOW-TO Guides for getting started with PHP applications.
XAMPP is meant only for development purposes. It has certain configuration settings that make it easy to develop locally but that are insecure if you want to have your installation accessible to others.
Start the XAMPP Control Panel to check the server status.
Community
XAMPP has been around for more than 10 years – there is a huge community behind it. You can get involved by joining our Forums, liking us on Facebook, or following our exploits on Twitter.



XAMPP → Apache → access.log → Webalizer → HTML analysis report

Result

Thus, web server log analysis using the Webalizer tool was studied. Apache was used to generate access logs, and the logs were examined to understand the information recorded during web requests. The Webalizer reporting process was studied to analyze server activity, including requests, accessed resources, traffic statistics, and HTTP response codes where available. The experiment demonstrated the usefulness of web server log analysis for monitoring and understanding website usage.